500Edge → Telegram Forwarder: Design Spec

Oct 16, 2026 · Greg Howard

Questions to answer before quoting

Please answer these with your quote. The answers decide how the build is done and what it costs.

1. **Dashboard: can you read the data directly, without simulating button clicks?** The data behind the "COPY SPX / ES" and "COPY SPY" buttons can probably be read from the page data or the network responses. Clicking buttons and reading the clipboard is more fragile and costs more to maintain. Please say which method you will use.
2. **Discord: how will you read #trade-alerts?** 500Edge declined bot access to its server. Reading the channel through a user account (a "self-bot") or by scraping the page likely breaks Discord's terms, and possibly 500Edge's, and could get the account banned. Please name the method and its risks.
3. **Dashboard: how often does it update during the session?** It showed "Updated 7:18 AM ET" pre-market. Please find out how often it changes intraday, and say how often you will check it.

Purpose and scope

The job is to **capture and forward**, exactly as written, two sources from 500Edge into Telegram. A separate program, which is not part of this job, reads the Telegram messages and does all of the interpretation.

- **Do not interpret the content.** Do not parse prices, strikes, or trade actions. Forward the exact text with the metadata in this spec.
- **Source A:** posts in the Discord #trade-alerts channel.
- **Source B:** the 500Edge web dashboard.
- **Two Telegram chats.** Alerts go to one chat and dashboard updates go to another, so each can be read and muted separately.
- **Times.** All timestamps are ISO 8601 with a UTC offset, for example `2026-10-16T09:53:12.345-04:00`.

Part A: Discord #trade-alerts

Every message posted in #trade-alerts is forwarded to the alerts chat as its own Telegram message. That includes edits and deletions.

Field	Requirement
Message ID	The Discord message ID (snowflake). Required on every message. It is used to match edits, deletions and replies, and to remove duplicates.
Timestamp	Taken from the message ID, to the millisecond, as ISO 8601 with a UTC offset. Do **not** use the displayed time (for example "9:53 AM"), which has no seconds and depends on the viewer's time zone.
Author	Display name, for example `Conner`.
Role	The author's role, for example `Head Trader`. Forward every author and tag the role. Do not filter.
Channel	Channel name, for example `trade-alerts`.

Field	Requirement
Event	`new`, `edited` or `deleted`.
Text	The full message text, exactly as posted, with line breaks kept. No trimming or reformatting.
Reply to	The message ID being replied to, if the message is a reply. Otherwise blank.
Images	The URL of each attached image. No OCR or image processing.

Rules

- **One Discord message = one Telegram message.** Discord groups consecutive posts from the same author under one header. Each underlying message must still be forwarded separately, with its own ID and timestamp. The sample alerts include cases like this (for example, three lines posted under one header on 7/30/26 at 10:04 AM).
- **Edits.** Forward the full new text, with the original message ID and `event=edited`.
- **Deletions.** Forward the message ID with `event=deleted`. No text is needed.
- **Long messages.** If a message is longer than Telegram's 4,096-character limit, split it into numbered parts that share the same message ID.

Part B: 500Edge dashboard

Check the dashboard regularly. Send an update to the dashboard chat **only when its "Updated" timestamp changes**. Each update contains the five items below.

#	Item	Source on the page	Format
1	SPX levels block	The text produced by the **COPY SPX / ES** button on the METRICS tab	The raw text, exactly as produced, with tabs and line breaks kept. Not a PDF or screenshot.
2	SPY levels block	The text produced by the **COPY SPY** button on the METRICS tab	Same as item 1.
3	Updated time	The "Updated:" line under the Dashboard title	ISO 8601 with a UTC offset. The copied text only carries the date, so this is required.
4	Extra metrics, SPX and SPY columns	The METRICS table	Reference Price; GEX (M) with its arrow (up, down or unchanged); 15-Minute, 1-Hour and 4-Hour Trend; Absolute Gamma Strike.
5	Calendar	The CALENDAR line under the Morning Note	The text exactly as shown, for example "No major releases", plus any HIGH IMPACT entries.

Optional: the Morning Note text, forwarded exactly as written with no parsing. Please quote it as a separate line item.

Reference: what the copied levels text looks like (SPX, 10/16/2026, shortened):

```
Copy SPX Metrics
10/16/2026
500Edge
SPX_PX  ES_PX   Level ID
7900    7948    SPX_CW
7776    7824    SPY_CW
7700    7748    G1
...
```

```
7754.15 7802.15 PREV_HIGH
7682 / 7806 7730 / 7854 IMP
```

The METRICS and LEVELS tabs produce the same text, so only the METRICS copy is needed. Do not convert, round or recompute any value.

Telegram message formats

Use these formats exactly. The receiving program depends on the header lines. Send everything as plain text with no Markdown or HTML formatting, so characters are not changed.

Alert message (to the alerts chat):

```
#ALERT
id=1296093345123456789
ts=2026-10-16T09:53:12.345-04:00
author=Conner
role=Head Trader
channel=trade-alerts
event=new
reply_to=
img=
part=1/1
---
Bot 7420 Calls at 9.00

This is a pre-FOMC trade and highly speculative.
```

- Header lines are `key=value`, one per line, in the order shown. Leave a value empty when it doesn't apply.
- `img` lists the image URLs, separated by spaces.
- `---` separates the header from the message text.
- A deleted message has `event=deleted` and no text after `---`.

Dashboard message (to the dashboard chat):

```
#DASH
updated=2026-10-16T07:18:00-04:00
SPX.ref=7744
SPY.ref=771.8
SPX.gex_m=195.0 up
SPY.gex_m=-795.0 up
SPX.trend_15m=Bullish
SPX.trend_1h=Bullish
SPX.trend_4h=Bullish
SPX.abs_gamma=7750
SPY.abs_gamma=770
calendar=No major releases
---SPX---
<raw COPY SPX / ES text>
---SPY---
<raw COPY SPY text>
```

- If the SPY column shows trend values, add `SPY.trend_15m`, `SPY.trend_1h` and `SPY.trend_4h`.
- If the Morning Note is in scope, add it at the end after a `---NOTE---` line.

Out of scope

Do not build or quote for the following.

- Parsing or interpreting alert text, such as prices, strikes, “1/2” or “Sold … avg”.
- OCR or any processing of images.
- Discord reactions, emojis, avatars, role icons and link previews.
- Other Discord channels, including general chat, voice-chat and market-updates.
- The full METRICS table beyond the fields in Part B, item 4.
- The LEVELS tab, and both “Levels” copy buttons, which duplicate METRICS.
- The 5-day implied move and range, and the 1-day implied move as a percentage.
- The SPX notional gamma chart, and the SPY, COMBO, VIX and GEX chart tabs.
- The scam banner, navigation buttons and disclaimer.
- Placing trades or connecting to any broker.

Speed, archive and reliability

Requirement	Target
Alert speed	Within 2 seconds of the Discord post, in at least 95% of messages, measured from the message ID timestamp to Telegram delivery.
Dashboard speed	Within 60 seconds of the “Updated” time changing.
Hours	8:00 AM to 4:15 PM ET on market days. Running all the time is fine.
No duplicates	The same message ID and event is sent only once, including after a restart.
No gaps	After a disconnect or restart, catch up on any missed messages, in order, before forwarding new ones.
Archive	Save every forwarded message, exactly as sent, in an append-only local file (JSON Lines, one file per day).
Error alerts	Send a short message to the dashboard chat if Discord or the dashboard can’t be reached for more than 60 seconds, and again when it recovers.
Credentials	Discord, dashboard and Telegram credentials are kept in a config file outside the code, and are never sent to Telegram or written to logs.

Acceptance tests and deliverables

The job is accepted when every test below passes during one full live market session.

Tests

- ☐ Every #trade-alerts message from the session appears in the alerts chat, one Telegram message each, with text matching Discord exactly.
- ☐ Messages posted together under one Discord header arrive as separate Telegram messages, each with its own ID and timestamp.
- ☐ Timestamps include milliseconds and the correct UTC offset, and match the message ID.
- ☐ An edited test message arrives with `event=edited` and the original ID. A deleted one arrives with `event=deleted`.
- ☐ A reply arrives with `reply_to` filled in. An image arrives with its URL.
- ☐ Alert speed meets the target, shown by a log comparing post time to delivery time.

- ☐ A restart during the session causes no duplicates and no missed messages.
- ☐ Dashboard updates arrive only when “Updated” changes, and contain all five Part B items.
- ☐ The SPX and SPY copied text matches what the buttons produce, character for character, with tabs kept.
- ☐ The archive file holds every message that was sent.

Deliverables

- Source code, with ownership transferred to Greg.
- A setup guide: install, config file, start, stop, and how to update credentials.
- One sample of each message type (new, edited, deleted, reply, image, dashboard), taken from a live session.
- The archive files from the acceptance session.